

Linux Real-World Assignment - TechNova Solutions

Scenario:

You are hired as a Linux System Administrator in a software company named **TechNova Solutions**. Your task is to prepare a Linux server environment for the company's developers and deploy a basic website securely. Students must complete all tasks using Linux commands only.

Task 1 - Create Project Structure

Create the following directory structure:

```
technova_project
├── website
├── backups
├── logs
├── scripts
└── developers_data
```

Requirements:

- Use Linux commands only
- Verify directories using `ls` command

```
fatima@fatima-VirtualBox:~$ sudo mkdir -p technova_project/{website,backup,logs,scripts,developers_data}
[sudo] password for fatima:
fatima@fatima-VirtualBox:~$ ls
Desktop  Documents  Downloads  hackerSetup.txt  Music  Pictures  Public  snap  technova_project  Templates  Videos
```

```
fatima@fatima-VirtualBox:~$ cd technova_project/
fatima@fatima-VirtualBox:~/technova_project$ ls
backup  developers_data  logs  scripts  website
```

Task 2 - Create Groups

Create the following groups:

```
frontend_team
backend_team
devops_team
```

```
fatima@fatima-VirtualBox:~/technova_project$ groupadd frontend_team
groupadd: Permission denied.
groupadd: cannot lock /etc/group; try again later.
fatima@fatima-VirtualBox:~/technova_project$ sudo groupadd frontend_team
fatima@fatima-VirtualBox:~/technova_project$ sudo groupadd backend_team
fatima@fatima-VirtualBox:~/technova_project$ sudo groupadd devops_team
```

Task 3 - Create Users

Create the following users and assign them to their groups:

ahmed → frontend_team

sana → backend_team

usman → devops_team

Requirements:

- Create home directories
- Set passwords
- Add users to correct groups

```
fatima@fatima-VirtualBox:~/technova_project$ sudo useradd -m ahmed
fatima@fatima-VirtualBox:~/technova_project$ sudo useradd -m sana
fatima@fatima-VirtualBox:~/technova_project$ sudo useradd -m usman
fatima@fatima-VirtualBox:~/technova_project$ sudo passwd ahmed
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
Sorry, passwords do not match.
New password:
Retype new password:
passwd: password updated successfully
fatima@fatima-VirtualBox:~/technova_project$ sudo passwd sana
New password:
Retype new password:
passwd: password updated successfully
fatima@fatima-VirtualBox:~/technova_project$ sudo passwd usman
New password:
Retype new password:
passwd: password updated successfully
```

Task 4 - Verify Users and Groups

Verify users and groups using:

```
id username
groups username
cat /etc/group
```

Students must explain:

- UID
- GID
- Supplementary Groups

```
fatima@fatima-VirtualBox:~/technova_project$ getent group frontend_team
frontend_team:x:1001:ahmed
fatima@fatima-VirtualBox:~/technova_project$ getent group backend_team
backend_team:x:1002:sana
fatima@fatima-VirtualBox:~/technova_project$ getent group devops_team
devops_team:x:1003:usman
fatima@fatima-VirtualBox:~/technova_project$ id ahmed
uid=1001(ahmed) gid=1004(ahmed) groups=1004(ahmed),1001(frontend_team)
fatima@fatima-VirtualBox:~/technova_project$ id sana
uid=1002(sana) gid=1005(sana) groups=1005(sana),1002(backend_team)
fatima@fatima-VirtualBox:~/technova_project$ id usman
uid=1003(usman) gid=1006(usman) groups=1006(usman),1003(devops_team)
```

Task 5 - Configure Permissions

Apply the following permissions:

```
website → 755
backups → 700
logs → 755
scripts → 744
```

Set Ownership:

```
website → ahmed:frontend_team
backups → root:root
logs → sana:backend_team
scripts → usman:devops_team
```

Verification Commands:

```
ls -ld
stat
```

```
fatima@fatima-VirtualBox:~/technova_project$ sudo chmod 755 website
fatima@fatima-VirtualBox:~/technova_project$ sudo chmod 700 backup
fatima@fatima-VirtualBox:~/technova_project$ sudo chmod 755 logs
fatima@fatima-VirtualBox:~/technova_project$ sudo chmod 744 scripts
fatima@fatima-VirtualBox:~/technova_project$ sudo chown ahmed:frontend_team website
fatima@fatima-VirtualBox:~/technova_project$ sudo chown root:root backup
fatima@fatima-VirtualBox:~/technova_project$ sudo chown sana:backend_team logs
fatima@fatima-VirtualBox:~/technova_project$ sudo chown usman:devops_team scripts
fatima@fatima-VirtualBox:~/technova_project$ stat website
  File: website
  Size: 4096          Blocks: 8          IO Block: 4096   directory
Device: 803h/2051d    Inode: 533723     Links: 2
Access: (0755/drwxr-xr-x)  Uid: ( 1001/   ahmed)   Gid: ( 1001/frontend_team)
Access: 2026-05-15 15:44:28.933480671 +0000
Modify: 2026-05-15 15:44:28.933480671 +0000
Change: 2026-05-15 17:45:17.506030158 +0000
 Birth: 2026-05-15 15:44:28.933480671 +0000
fatima@fatima-VirtualBox:~/technova_project$ ls -ld website
drwxr-xr-x 2 ahmed frontend_team 4096 May 15 15:44 website
fatima@fatima-VirtualBox:~/technova_project$ ls -ld backup
drwx----- 2 root root 4096 May 15 15:44 backup
fatima@fatima-VirtualBox:~/technova_project$ ls -ld logs
drwxr-xr-x 2 sana backend_team 4096 May 15 15:44 logs
fatima@fatima-VirtualBox:~/technova_project$ ls -ld scripts
drwxr--r-- 2 usman devops_team 4096 May 15 15:44 scripts
```

Task 6 - Install Apache Web Server

Install Apache web server.

Requirements:

- Install Apache
- Start Apache service
- Enable Apache service
- Verify Apache status

Verification Command:

`systemctl status apache2`

```
fatima@fatima-VirtualBox:~/technova_project$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.52-1ubuntu4.20).
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
fatima@fatima-VirtualBox:~/technova_project$ sudo systemctl start apache2
fatima@fatima-VirtualBox:~/technova_project$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-in
stall.
Executing: /lib/systemd/systemd-sysv-install enable apache2
fatima@fatima-VirtualBox:~/technova_project$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese
   Active: active (running) since Fri 2026-05-15 14:57:40 GMT; 3h 27min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 871 (apache2)
    Tasks: 55 (limit: 9428)
   Memory: 7.7M
      CPU: 1.951s
   CGroup: /system.slice/apache2.service
           └─871 /usr/sbin/apache2 -k start
             └─873 /usr/sbin/apache2 -k start
               └─874 /usr/sbin/apache2 -k start

May 15 14:57:34 fatima-VirtualBox systemd[1]: Starting The Apache HTTP Server...
```

Task 7 - Deploy Website

Create index.html file inside website folder.

Website must contain:

- Company Name
- Welcome Message
- Student Name

Example:

```
<h1>Welcome to TechNova Solutions</h1>
<p>Managed by Linux Students</p>
```

```
fatima@fatima-VirtualBox:~/technova_project$ cd website/
fatima@fatima-VirtualBox:~/technova_project/website$ ls
fatima@fatima-VirtualBox:~/technova_project/website$ sudo touch index.html
fatima@fatima-VirtualBox:~/technova_project/website$ ls
index.html
fatima@fatima-VirtualBox:~/technova_project/website$ sudo nano index.html
fatima@fatima-VirtualBox:~/technova_project/website$ cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>TechNova Solutions</title>
</head>
<body>
  <h1>Welcome to TechNova Solutions</h1>
  <p>Managed by Devops Student: Fatima Sarwar</p>
  <p>This server environment is ready for production.</p>
</body>
</html>
```

Task 8 - Network Verification

Perform network verification using:

```
ip a
ping google.com
hostnamectl
```

Students must explain:

- IP Address
- Network Interface
- Packet transmission

```
fatima@fatima-VirtualBox:~/technova_project/website$ ping google.com -c4
PING google.com (142.250.202.238) 56(84) bytes of data:
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=1 ttl=255 time=19.8 ms
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=2 ttl=255 time=20.5 ms
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=3 ttl=255 time=20.6 ms
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=4 ttl=255 time=20.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3040ms
rtt min/avg/max/mdev = 19.784/20.334/20.552/0.319 ms
fatima@fatima-VirtualBox:~/technova_project/website$ hostnamectl
Static hostname: fatima-VirtualBox
    Icon name: computer-vm
    Chassis: vm
    Machine ID: 4fbdbef25d4f4ec09adc333b3cc59e5d
    Boot ID: 2787658df9954122932e5eda0d25e55c
    Virtualization: oracle
Operating System: Ubuntu 22.04.5 LTS
    Kernel: Linux 6.8.0-111-generic
    Architecture: x86_64
Hardware Vendor: innotek GmbH
```

Task 9 - System Monitoring

Monitor system using:

```
top
free -m df -h uptime
```

Answer the following:

1. How much RAM is available?
2. What is the disk usage?
3. What is the system uptime

```
fatima@fatima-VirtualBox:~/technova_project/website$ free -m
              total        used        free      shared  buff/cache   available
Mem:           7941         1512         3992           54         2436         6227
Swap:          2047           0         2047

fatima@fatima-VirtualBox:~/technova_project/website$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            795M  2.1M  793M   1% /run
/dev/sda3        24G   14G   9.1G  60% /
tmpfs            3.9G     0   3.9G   0% /dev/shm
tmpfs            5.0M  4.0K  5.0M   1% /run/lock
/dev/sda2        512M  6.1M  506M   2% /boot/efi
tmpfs            795M  112K  795M   1% /run/user/1000

fatima@fatima-VirtualBox:~/technova_project/website$ uptime
18:46:11 up  2:59,  1 user,  load average: 0.01, 0.13, 0.18
```

Bonus Task - Backup Project

Create backup archive:

```
tar -czvf tech_backup.tar.gz technova_project/
```

```
fatima@fatima-VirtualBox:~$ tar -czvf tech_backup.tar.gz technova_project/
technova_project/
tar: technova_project/backup: Cannot open: Permission denied
technova_project/logs/
technova_project/website/
technova_project/website/index.html
technova_project/developers_data/
technova_project/scripts/
tar: Exiting with failure status due to previous errors
fatima@fatima-VirtualBox:~$ sudo tar -czvf tech_backup.tar.gz technova_project/
[sudo] password for fatima:
technova_project/
technova_project/backup/
technova_project/logs/
technova_project/website/
technova_project/website/index.html
technova_project/developers_data/
technova_project/scripts/
fatima@fatima-VirtualBox:~$ ls
Desktop      Downloads    Music        Public       tech_backup.tar.gz  Templates
Documents    hackerSetup.txt  Pictures     snap         technova_project    Videos
```